

C. auris SCF1 RNA-seq Reproduction Study

Reproducing Santana & O'Meara 2023 (Science 381:1461) using B8441 V3 assembly (GCA_002759435.3)

Study Overview

This analysis reproduces the core RNA-seq experiment from Santana & O'Meara 2023 (Science 381:1461-1467), which identified SCF1 (Surface Colonization Factor) as the principal adhesin driving *C. auris* surface colonization, skin infection, and virulence. The paper used the V2 B8441 genome; we use the current V3.

****Dataset:**** BioProject PRINA904261 — 6 paired-end FASTQ samples (NextSeq 2000, 2×50 bp, ~22–30 M reads)

****Conditions:**** AR0382 WT (n=2) · AR0382 tnSWI1 (n=2) · AR0387 (n=2) — all grown to log phase in YPD

****Library:**** Illumina Stranded Total RNA Prep with Ligation + Ribo-Zero Plus → reverse-stranded

****Reference:**** GCA_002759435.3 · B8441 V3 · chromosome-level · 7 chr · 5,424 genes · 12.4 Mb

****Platform:**** usegalaxy.org | IWC rnaseq-pe + rnaseq-de workflows

Methods Comparison

Step	Paper (Santana 2023)	This Analysis
Trimming	Cutadapt	fastp (IWC rnaseq-pe)
Alignment	STAR (ENCODE parameters)	STAR (ENCODE parameters)
Counting	featureCounts -s 2	featureCounts -s 2 (reverse)
DE testing	DESeq2	DESeq2 (IWC rnaseq-de)
Reference	GCA_002759435.2 / V2	GCA_002759435.3 / V3
Annotation	FungiDB	NCBI RefSeq GTF (V3)
ID mapping	Direct (V2 IDs)	Diamond RBH V3↔V2 (4,567 pairs)
Platform	Galaxy (usegalaxy.org)	Galaxy (usegalaxy.org)